

Calcul constantes cosmologiques

Célérité : c dans le vide $c = \frac{1}{\sqrt{\mu\epsilon}} = 299792458 \text{ m/s}$

```
In [14]: %display latex
c=299792458;mu=n(pi)*4E-7;ep=1/(mu*c^2)
c=1/sqrt(mu*ep)
show(LatexExpr(r"c=\frac{1}{\sqrt{\mu \epsilon}}"),
      round(c)," m/s")
```

$$c = \frac{1}{\sqrt{\mu\epsilon}} = 299792458 \text{ m/s}$$

Constante cosmologique $\Lambda = \frac{8\pi G\rho_\Lambda}{c^2}$

```
In [27]: G=6.67384E-11;
La=(8*3.1415*G*2.232E-27)/c^2;La
show(LatexExpr(r"\Lambda=\frac{8\pi G \rho_\Lambda}{c^2}"),La,LatexExpr(r"m^{-2}"))
```

$$\Lambda = \frac{8\pi G\rho_\Lambda}{c^2} = 4.16538957764143 \times 10^{-53} \text{ m}^{-2}$$

Longueur de Planck $L_p = 1,61622938 \times 10^{-35} \text{ m}$

```
In [33]: h=6.626E-34;hb=h/(2*n(pi));hb=h/(n(pi)*2)
Lp=sqrt(hb*G/c^3)
show(LatexExpr(r"L_p=ct_p=\sqrt{\frac{\hbar G}{c^3}}"),Lp,LatexExpr(r"m"))
```

$$L_p = ct_p = \sqrt{\frac{\hbar G}{c^3}} = 1.61619077108993 \times 10^{-35} \text{ m}$$

Masse apparente de l'univers m_u

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In [41]: a=7.297735256698E-3;B=3-sqrt(5.0);Re=2.817940E-15;
H0=(c*(a^19)*(sqrt(B)))/Re;Mu=c^3/(G*H0)
show(LatexExpr(r"L_p=ct_p=\sqrt{\frac{\hbar G}{c^3}}"),Lp,LatexExpr(r"m"))
```

$$L_p = ct_p = \sqrt{\frac{\hbar G}{c^3}} = 1.61619077108993 \times 10^{-35} \text{ m}$$

In []: